Niti B. Jadeja, Ph.D.

E-mail : nitijadeja22@gmail.com Mobile: +1 (703) 419-0938

Research Profile

Microbiologist with 11+ years of research in infection biology, innate immunity and genomics. Current work uses spatiotemporal analysis of *Shigella* infection to study macrophage and neutrophil subsets and IFN-γ and NF-κB signaling at invasion sites. Contributes to a collaboration with icddr,b, Bangladesh, on tissue remodeling in human biopsies, and mentors undergraduate researchers.

Technical Expertise

- **Infection models and tissue:** infant rabbit colon infection model; NAIP-NLRC4-deficient mice, human colon biopsies; *S. flexneri* infection studies
- **Spatial and imaging methods:** 100+ genes with RNAscope combined with immunofluorescence (RNA-protein co-detection) for epithelial and bacterial markers; image quantification in QuPath and CellProfiler
- **Sequencing and genomics:** RNA-seq and proteomics data analysis; integration with GTEx (dbGaP-approved controlled access); Oxford Nanopore MinION and Illumina sequencing; metagenomics
- **Innate immunology:** macrophage and neutrophil subset profiling; IFN-γ and NF-κB pathway analysis; neutrophil-bacteria assays

Education

Ph.D. in Biotechnology, RTMNU, Nagpur, India (2019)
Doctoral research on the topic entitled 'Screening of effluent treatment plant metagenome for novel oxygenase' (Research carried out at India's National Environmental Engineering Research Institute)

M.Sc. in Biotechnology, B.R. Doshi School of Biosciences, Sardar Patel University, India (2010)

B.Sc. in Botany, Sir P.T. Science College, Veer Narmad South Gujarat University, India (2008)

Professional Experience

Post-Doctoral Research Associate, Agaisse Lab, Microbiology-MIC Dept., School of Medicine, University of Virginia, since May 2022.

- Designed and led a time-course study of *Shigella* invasion in the infant-rabbit colon, identifying stage-specific macrophage and neutrophil populations and mapping localized disruption of IFN-γ and NF-κB signaling near invasion sites, using RNAscope combined with immunostaining for epithelial cells and bacteria, and quantitative tissue imaging. (bioRxiv 2026)
- Led RNA-seq analyses of *S. flexneri* and *S. sonnei* infection in human colon biopsies and infant-rabbit tissue; assessed batch effects and compared infection-associated expression with GTEx colon data. Also analyzed human biopsy RNA-seq, proteomic and spatial expression data on tissue-remodeling markers in *Shigella* pathogenesis. (manuscript in preparation)
- Conducted a *Shigella* infection time course in NAIP-NLRC4-deficient mice to compare host-response dynamics with the infant-rabbit model. Performed breeding crosses with a septin-altered line and managed colony maintenance with the UVA collaborator's lab as groundwork for planned live imaging.
- Identified and functionally validated MxiE-box motifs upstream of *yccE* and *yfdF*, showing that targeted motif mutations abolished promoter activation under type 3 secretion-inducing conditions; complementation restored activation from intact promoters. *(mSphere, 2022)*

Post-Doctoral Fellow at Ashoka Trust for Research in Ecology and the Environment, Bangalore, India, May 2020 - May 2022, for 'The National Mission on Biodiversity and Human Well Being' (phase 1-2) 2020-2021.

Research Scientist at The Foundation for Medical Research, Mumbai, India from September 2018 - September 2019.

CSIR-Senior Research Fellow at the Environmental Biotechnology and Genomics Division, NEERI, Nagpur from October 2015 - September 2018.

Mobility grant by Academy of Finland to visit SYKE, Finnish Environment Institute, to work on quantitative and metagenomic analysis of Baltic Sea samples (September-November 2016).

Project Fellow in the 12th Five-Year Plan Project, CSIR-India, at the Genomics Division, NEERI, Nagpur and performing doctoral research on the topic 'Screening the Effluent Treatment Plant Metagenome for Novel Oxygenase' from March 2013 - July 2015.

Project Fellow in the project entitled 'Metagenome analysis for metabolic pathways present in activated biomass at common effluent treatment plant (CETP)' from August 2010 - February 2013.

Awards/Fellowships

- EMBO Travel grant for EMBO Workshop *Shigella*: from biology to prevention, Paris, April-2026
- EMBO Travel grant When biology of endocytosis meets physics, Venice, September-2023
- OFID Travel Grant for IWA-Water Congress at Sri Lanka, December-2019
- DST Young Scientist travel award, Leipzig, August-2018
- International Water Association travel award, April-2018
- CSIR travel grant to South Africa, Johannesburg, March-2018
- Best presentation award at Women in Science without Borders conference, Johannesburg, March 2018.
- Academy of Finland visiting researcher grant, September-November 2016.
- Best poster award at National Conference, Meedta, at HNB Garhwal University, October 2015.
- India's National Graduate Aptitude Test in Engineering (GATE) Biotechnology-2011 qualified.
- CSIR-Senior Research Fellowship (October 2015-September 2018)

Teaching and Advising Experience

- Tomorrow's Professor Today program, Center for Teaching Excellence, University of Virginia (2026 - present).
- Mentored undergraduate researcher Rishab Rao Katikanani in study design and de novo bioinformatics analysis, from planning through data interpretation (2025).
- Mentored undergraduate researcher Sarayu Devbhuktani in sequencing a laboratory *Shigella* strain using Oxford Nanopore MinION technology (2024).
- Guided master's dissertation research of more than 10 students (2010-2017).
- Lessons in wastewater management, Academy of Scientific and Innovative Research (AcSIR), CSIR-NEERI (2013-2014).

Certifications and Training

- Animal Research Training (AALAS Learning Library, University of Virginia; 7 CEUs total)
- UVA Orientation to the Animal Care and Use Program (2022)
- Introduction to Rabbits (2022)
- Working with the Laboratory Mouse (2023)
- UVA Refresher for Animal Handlers (2025)

Other Certifications

- Certificate of Completion, Immuno-oncology (HMX), Harvard Medical School (2026)

Service and Membership

- Vice Chair, Professional Development Committee, UVA-Postdoctoral Association (2023-2026)
- Founder MIC Department Postdoctoral Leadership (2023-2024).
- Young Water Professionals Chapter Coordinator, IWA Global Emerging Water Leaders Steering Committee (2020-2022).
- Organizer, 5th Global Health Foundation meeting, Mumbai Zonal Centre, CSIR-NEERI, Nehru Science Centre, Mumbai (2018)
- Lifetime Member, SEDRIC (Wellcome Trust, UK); Member, OWSD Asia Chapter (ID 6243)

Policy Contributions

- Contributions to the first order draft of Working Group I of the IPCC Sixth Assessment Report (AR6), June 2019.
- Contributions to the draft terms for the 'Independent Panel on Evidence for Action against Antimicrobial Resistance' of the UN Interagency Coordination Group (IACG) on Antimicrobial Resistance, June 2020.

Publications

- Jadeja NB, Devasure BA, Yum LK, Agaisse HF (2026). Sequential invasion of epithelial and immune cells in shigellosis. bioRxiv. doi.org/10.64898/2026.09.11.751053
- Hall, C. P., Jadeja, N. B., Sebeck, N., & Agaisse, H. (2022). Characterization of MxiE- and H-NS-Dependent Expression of ipaH7.8, ospC1, yccE, and yfdF in *Shigella* flexneri. mSphere, 7(6), e00485-22.
- Jadeja, N. B., & Worrich, A. (2022). From gut to mud: dissemination of antimicrobial resistance between animal and agricultural niches. Environmental Microbiology, 24(8), 3290-3306.
- Kumar, A., Nath, S. S., Sridhar Sudarshan, A., Iyer, V., Jadeja, N. B., Panchamia, N., ... & Tatu, U. (2022). Genome Sequences of Five Indian Canine Rabies Virus Isolates Obtained Using Oxford Nanopore Technologies Sequencing. Microbiology Resource Announcements, 11(5), e01246-21.
- Paul, M. M., Jadeja, N. B., Sidhu, N., & Vanak, A. T. (2020). Whither One Health in India? National Institute of Advanced Studies, Bangalore, 55, 40.
- Yadav, T. C., Jadeja, N. B., & Kapley, A. (2020). Metagenomic Insights in Activated Biomass Treating Industrial Wastewater at Different DO Levels. Applied Biochemistry and Biotechnology.
- Bhardwaj, P., Singh, K. R., Jadeja, N. B., Phale, P. S., & Kapley, A. (2020). Atrazine Bioremediation and Its Influence on Soil Microbial Diversity by Metagenomics Analysis. Indian Journal of Microbiology, 1-4.
- Jadeja, N. B., Purohit, H. J., & Kapley, A. (2019). Decoding microbial community intelligence through metagenomics for efficient wastewater treatment. Functional & Integrative Genomics, 19, 839-851.
- Jadeja, N. B., Moharir, P., & Kapley, A. (2019). Genome Sequencing and Analysis of Strains Bacillus sp. AKBS9 and Acinetobacter sp. AKBS16 for Biosurfactant Production and Bioremediation. Applied Biochemistry and Biotechnology, 1-13.
- Sahu, N., Sharma, G., Chandrashekhar, B., Jadeja, N. B., Kapley, A., Pandey, R. A., & Sharma, A. (2019). Performance evaluation of methanogenic digester using kitchen waste for validation of optimized hydrolysis conditions for reduction in ammonia accumulation. Renewable Energy, 139, 110-119.
- Rudrashetti, A. P., Jadeja, N. B., Gandhi, D., Juwarkar, A. A., Sharma, A., Kapley, A., & Pandey, R. A. (2017). Microbial population shift caused by sulfamethoxazole in engineered-Soil Aquifer Treatment (eSAT) system. World Journal of Microbiology and Biotechnology, 33, 121.
- Chattaraj, S., Purohit, H. J., Sharma, A., Jadeja, N. B., & Madamwar, D. (2016). Treatment of Common Effluent Treatment Plant Wastewater in a Sequential Anoxic–Oxic Batch Reactor by Developed Bacterial Consortium VN11. Applied Biochemistry and Biotechnology, 179, 514-529.

- Yadav, T. C., Pal, R. R., Shastri, S., Jadeja, N. B., & Kapley, A. (2015). Comparative metagenomics demonstrating different degradative capacity of activated biomass treating hydrocarbon contaminated wastewater. Bioresource Technology, 188, 24-32.
- Kapley, A., Liu, R., Jadeja, N. B., Zhang, Y., Yang, M., & Purohit, H. J. (2015). Shifts in Microbial Community and Its Correlation with Degradative Efficiency in a Wastewater Treatment Plant. Applied Biochemistry and Biotechnology, 176, 2131-2143.
- Jadeja, N. B., More, R. P., Purohit, H. J., & Kapley, A. (2014). Metagenomic analysis of oxygenases from activated sludge. Bioresource Technology, 165, 250-256.

Published Proceedings

- Jadeja, N. B., Sahu, N., Pandey R. A., Kapley A. (2019): Microbiome analysis to understand the biotransformation reactions responsible for conversion of food waste to energy in Biogas reactors at the IV. CMP International Conference on Monitoring & Process Control of Anaerobic Metabolisms at Leipzig, Germany, 26-27 March 2019, 35-36.
- Jadeja, N. B., Kapley A. (2018): Can the black box be cracked? High throughput sequencing to understand wastewater bioremediation. Women in Science Without Borders (WiSWB) Conference at Johannesburg, South Africa, March 2018, 11-12.
- Jadeja, N. B., Kapley A. (2018): Decoding microbial intelligence through metagenomic analysis to redefine biological wastewaters treatment process. 19th UK IWA Young Water Professionals Conference at Cranfield University, Cranfield, UK, April 2018, 123-125.

Book Chapters

- Jadeja, N. B., & Tamim Vanak, A. (2023). High-Throughput Techniques to Understand Evolution and Transmission Trends of Rabies Virus in Asian Countries. In One Health for Dog-mediated Rabies Elimination in Asia: A Collection of Local Experiences (pp. 220-234). GB: CABI.
- Jadeja, N. B., & Kapley, A. (2023). Designing knowledge-based bioremediation strategies using metagenomics. In Metagenomic Data Analysis (pp. 195-208). New York, NY: Springer US.
- Yadav, S., Jadeja, N. B., Dafale, N. A., Purohit, H. J., & Kapley, A. (2019). Pharmaceuticals and personal care products mediated antimicrobial resistance: future challenges. In Pharmaceuticals and Personal Care Products: Waste Management and Treatment Technology (pp. 409-428). Butterworth-Heinemann.
- Jadeja N. B., Yadav S., & Kapley A. (2018). Applications of metagenomics approaches in bioremediation of environmental pollutants. In Recent Advances in Environmental Management. CRC Press, Taylor & Francis Group, USA.
- Kapley A., Jadeja, N. B., Paliwal V., Yadav T. C., & Purohit H. J. (2015). Microbial Genomics and Bioremediation of Industrial Wastewater. In Chandra R. (ed.), Microbial Genomics and Bioremediation of Industrial Wastewater. U.K., Taylor and Francis.

Conferences & Presentations

- Neutrophil Conference, San Diego, 21-23 September 2026 (poster).
- EMBO Workshop *Shigella*: from biology to prevention, Paris, 20-24 April 2026 (short talk).
- International Union of Immunological Societies, Vienna, Austria, 17-22 August 2025 (talk).
- The Neutrophil | International Symposium, Munich, Germany, September 2024
- EMBO's When biology of endocytosis meets physics, San Servolo, Venice, Italy, 2023.
- Surveillance and Epidemiology of Drug-resistant Infections Consortium (SEDRIC) Global Meeting, Wellcome Trust, 9 October 2019, Euston, London (Talk).

- Review of Progress on Antimicrobial Resistance Global meet, 8th October 2019, Chatham House, Royal Institute of International Affairs, St James's Square, London (Panel discussion).
- Speaker at the IV. CMP International Conference on Monitoring & Process Control of Anaerobic Metabolisms, 26-27th March 2019, Leipzig, Germany (Talk).
- 17th International Symposium on Microbial Ecology-ISME17, Leipzig, Germany, 12-17th August 2018 (poster presentation).
- International Water Association-19th Young Professional conference, Cranfield University, United Kingdom, 16-18th April 2018 (Flash presentation & poster presentation)
- Second International Women in Science without borders (WISWB-2018), University of Johannesburg, South Africa, 21-23 March 2018 (Oral presentation, Session chair)

Academic Achievements

- Gold Medallist in Graduation in Botany in the year 2008.
- National Cadet Corps 'C' Certificate holder with the rank of Senior Under Officer.

Profile Links

- Researchgate: https://www.researchgate.net/profile/Niti_Jadeja
- Scholar: https://scholar.google.co.in/citations?user=OOuxVm4AAAAJ&hl=en&oi=ao
- ORCID: https://orcid.org/0000-0002-2199-9605
- Scopus Author ID: 56060693800
- ResearcherID: G-3539-2010